Anjaneya Bhardwaj

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INTRODUCTION

As an aspiring AI/ML Engineer, I focus on applying Agentic AI, Reinforcement Learning and Deep Learning and to solve complex, real-world problems. My experience includes designing diverse systems, ranging from multi-modal classification models and generative diffusion architectures for medical time-series forecasting to developing multi agentic systems. I am committed to leveraging my skills to develop impactful, data-driven solutions within a collaborative environment.

EDUCATION

University at Buffalo, SUNY **2025 - 2026**

MS, Artificial Intelligence

- Developed expertise in Generative Diffusion Models, Reinforcement Learning, and Neural Network design. Led collaborative teams to build and optimize predictive models, managing high-performance data workflows and scaling neural architectures.
- **Coursework:** Data Intensive Computing, Numerical Methods for Data Science, Reinforcement Learning, Computer Vision, Pattern Recognition, Basics of AI, Intro to ML

Jaypee Institute of Information Technology (JIIT) **2020 - 2024**

B.Tech, Information Technology

- Developed skills in problem solving, leadership, time management, product planning, solution delivery, and cross-team collaboration over four years of engineering.
- **Coursework:** Data Mining, Web Algorithms, Information Retrieval, Machine Learning, Big Data, Financial Accounting, Operations Research

EXPERIENCE

Globant X University at Buffalo **Jan 2026 - May 2026**

AI Engineer – Team Lead

- Engineered an end-to-end **AI procurement optimization system** for a Mexican fuel station network operating 617 stations with a ~$189B MXN ($9.5B USD) annual procurement baseline, targeting 5–15% cost reduction through systematic supplier selection. And coordinated a team of 3 on testing, data validation and UI integration.
- Designed and built a **Multi-Agent AI system** with 1 orchestrator using 7 local tools and 3 live web-search sub-agents running in parallel(weather, route security, and fuel market intelligence), reducing risk assessment latency by **~57% (210s → 90s),** using **LangChain** and **LangGraph**.
- Developed a **4-stage ingestion pipeline** (Format Reader → Extractors → Validator → Normalizer) processing 7,251 supplier price files across 4 formats (PDF, HTML, XLSX, TXT) and 4 suppliers spanning a 2-year price history.
- Built a **Leakage Score model** (0–100) quantifying supplier delivery risk from 252,117 historical orders. Modelled as a weighted composite of late delivery rate (40%), lead time variance (35%), and distance risk (25%). Implemented a **multi-criteria decision engine (AHP, TOPSIS, PROMETHEE II)** and a **MILP order-splitting optimizer** enforcing regulatory supplier floor constraints, grade-specific MOQs, and station-level tanker capacity limits.
- Delivered a production-ready Streamlit dashboard with 4 modules: document ingestion, AI-assisted procurement recommendations, analytics (KPI strip, spend trends, supplier mix, days-of-stock), and a conversational procurement agent that can converse in both Spanish and English.

ECHO India **Jun 2023 - Jul 2023**

Data Science Intern

- Collaborated with senior data scientists and analysts to collect, clean, and preprocess datasets from diverse sources using **Python and pandas**, enabling downstream predictive modeling
- Created visualizations to illustrate key trends in user behavior, helping stakeholders quickly identify areas for improvement
- Performed **NLP** tasks and built a sentiment analyzer with the **VADER** lexicon in Python, automatically classifying user feedback and highlighting positive and negative trends for the product team.

PROJECTS

Multi-Agent RL for Network Intrusion Detection | GitHub

- Developed an AI-driven cybersecurity defense system utilizing **Multi-Agent Reinforcement Learning (MARL)** to protect critical network assets under resource constraints.
- Modeled enterprise networks as graphs using **GNNs** to make agents topology-aware, achieving a **47% improvement** over baseline metrics using **Proximal Policy Optimization (PPO)**

Exploring Sepsis detection using Diffusion Models | GitHub

- Engineered a **Conditional Denoising Diffusion Probabilistic Model (DDPM)** for medical time-series, treating vital sign forecasting as a generative "re-painting" task to visualize outcome distributions providing **superior clinical utility over traditional LSTMs** by providing a cloud of outcomes.
- Implemented a 1D ResNet-style architecture in PyTorch to process temporal dependencies and encode diffusion time-steps via sinusoidal embeddings

Healthcare Data Management and Warehousing System | GitHub

- Designed a normalized PostgreSQL schema (3NF) and a Star Schema data warehouse optimized for complex analytical queries and high-performance reporting.
- Built an automated pipeline using Python and **dbt** to engineer Fact/Dimension tables for hospital revenue and patient workload metrics.
- **Containerized** the full stack (Database, API, and Pipeline) using **Docker and Docker Compose**, ensuring consistent development and deployment across environments.
- Implemented performance tuning and database indexing, reducing execution time from **616ms to 53ms** (approx. 90% improvement)

Deafine - Real-time Speaker Detection | GitHub

- A simple Python accessibility tool that provides real-time speaker detection and live transcription using **ElevenLabs Speech-To-Text API** for deaf and hard-of-hearing users.
- Implemented real time audio processing with **WebSocket** and made sure that there were no redundant API calls when there is silence through batched API calls.
- Secured the Best Use of ElevenLabs Award at **MLH-UB Hackathon**

Multi Modal Genre Prediction | GitHub

- Engineered a fusion model integrating textual plot summaries (via **LSTM**) and visual movie posters (via **MLP**) to enhance classification accuracy on a dataset of **100,000+ movies**.
- Implemented custom data augmentation and undersampling techniques to solve class imbalance across six main genres, **resulting in a 72% accuracy rate** for the integrated **multi-modal system**.

CERTIFICATIONS

- **Supervised Machine Learning- Regression and Classification: Issued Nov 2024**. Learned about regression for predictive numerical analysis and classification for accurate data categorization for robust decision-making in diverse applications. Online Course by Stanford, DeepLearning.AI.
- **Affective Computing: Issued May 2024.** Gained skills in emotion recognition, computational modeling of emotions, and analysis of multimodal data (voice, facial expressions, physiological signals) using machine learning and signal processing techniques. Developed an understanding of ethical, legal, and social implications in human-machine interaction. Online Course by NPTEL, IIIT-D.

LEADERSHIP AND ACTIVITIES

Light De Literacy NGO **2020 - 2024**
Admissions Head

- Served as Admissions Head, facilitating the entry process for underprivileged children and coordinating large-scale fundraising and donation drives.

Progno.AI Startup Initiative **2023**
Team Lead

- Led a team to design a platform for early detection of lifestyle diseases, ensuring continuity of care and improved patient experiences.

SKILLS

- **AI/Data**: PyTorch, TensorFlow, LLMs, Reinforcement Learning, Neural Networks, PostgreSQL, Pandas, NLP, Copilot, dbt, Generative AI, Diffusion Models, GNN, Computer Vision, RAG, Scikit Learn, LLM, Deep Learning, Prompt Engineering, AI Agents, Multi-Agent Orchestration, LangChain, LangGraph Langsmith
- **Languages**: Python, C++
- **Infrastructure**: Docker, Docker Compose, Kubernetes, REST APIs (FastAPI), Git, ETL, Spark
- **Analytical**: Statistical Modeling, Data Visualization (Plotly, Seaborn), Data Manipulation, Data Mining, Data Pipelines, Performance Tuning, Multi Criteria Decision Modeling.